

Introduction to FX & FX Derivatives

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Part I

FX Fundamentals

1. What is Foreign Exchange (FX)?

- ▶ Foreign Exchange (FX/Forex) is the exchange of one currency for another
- ▶ Every FX transaction involves a currency pair — e.g., EUR/USD, USD/JPY, GBP/CHF
- ▶ The base currency (first) is bought/sold; the quote currency (second) is the price
- ▶ Exchange rate represents the price of 1 unit of base currency in quote currency terms
- ▶ FX is the backbone of global trade, investment, tourism, and capital flows
- ▶ Unlike equities, FX is always a relative value trade — you're long one currency and short another

Reading a Quote

EUR / USD = 1.0850

EUR = Base currency

USD = Quote currency

→ 1 EUR costs 1.0850 USD

Bid: 1.0848 (dealer buys EUR)

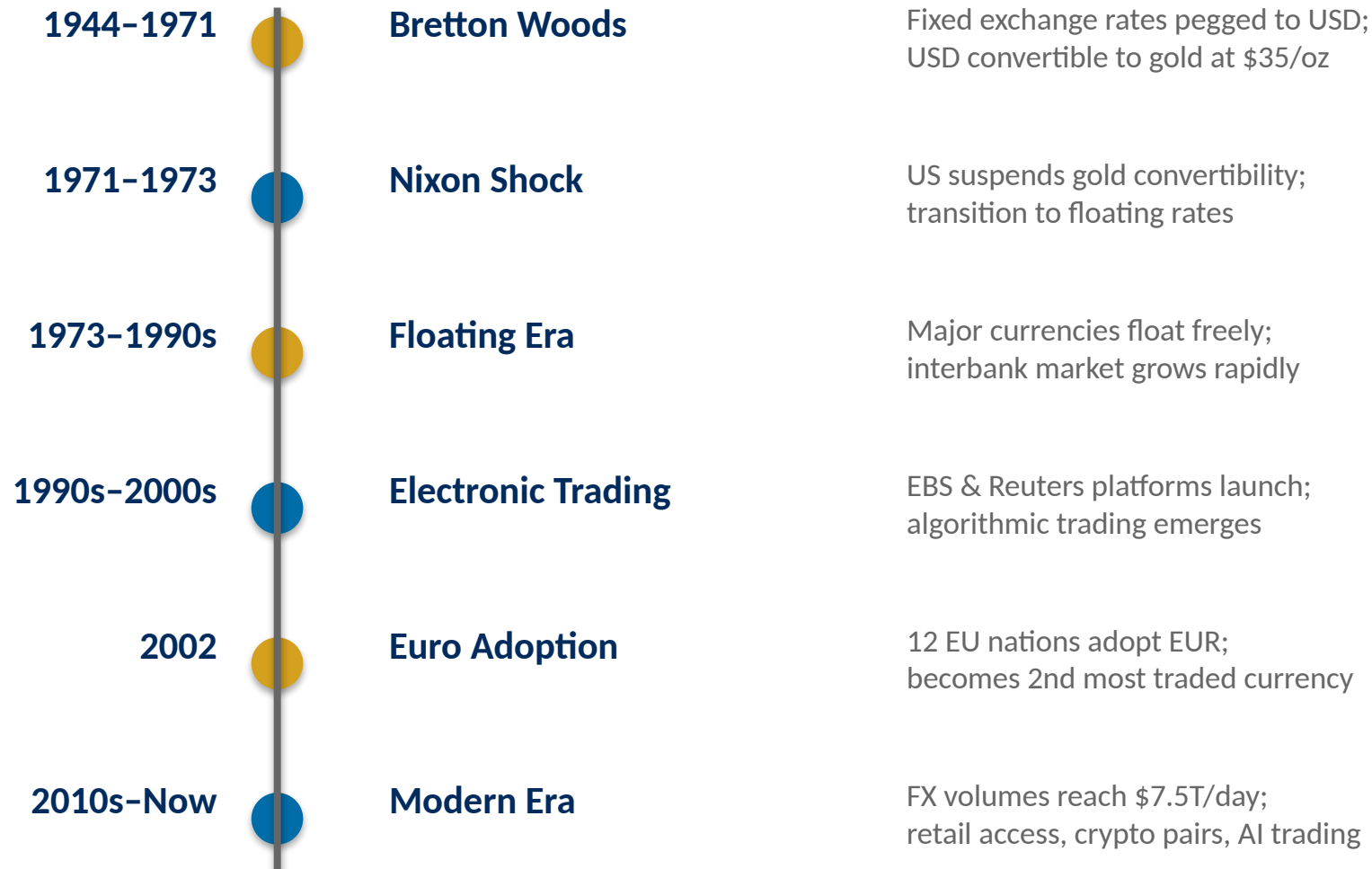
Ask: 1.0852 (dealer sells EUR)

Spread: 0.0004 = 4 pips

1 pip = 0.0001 for most pairs

1 pip = 0.01 for JPY pairs

2. History & Evolution of FX Markets



3. The Global FX Market — Size & Structure

- ▶ The FX market is the largest and most liquid financial market globally
- ▶ Average daily turnover: ~\$7.5 trillion (BIS Triennial Survey, 2022)
- ▶ Operates 24 hours a day, 5½ days a week — follows the sun across time zones
- ▶ Decentralised OTC market — no single exchange or clearing house (though CLS settles ~\$6T/day)
- ▶ Trading sessions: Sydney → Tokyo → London → New York (overlap = highest liquidity)

4. Market Participants & Their Roles

Central Banks

- ▶ Implement monetary policy through FX intervention
- ▶ Manage foreign exchange reserves
- ▶ Defend or adjust exchange rate targets
- ▶ Examples: Fed, ECB, BOJ, PBOC

Commercial / Investment Banks

- ▶ Act as market makers — quote bid/ask
- ▶ Facilitate client trades (corporates, funds)
- ▶ Proprietary trading and risk warehousing
- ▶ Top dealers: JPM, Citi, UBS, Deutsche

Corporates

- ▶ Hedge commercial FX exposures (imports/exports)
- ▶ Manage foreign subsidiary cash flows
- ▶ Treasury operations and cash pooling
- ▶ Natural hedging + derivative overlays

Asset Managers & Hedge Funds

- ▶ Currency overlay programmes for portfolio hedging
- ▶ Macro / carry / momentum FX strategies
- ▶ Alpha generation through FX derivatives
- ▶ Pension funds, sovereign wealth funds, CTAs

Retail Traders & Brokers

- ▶ Access via online platforms (leveraged margin)
- ▶ Represent ~5-7% of total FX volume
- ▶ Typically short-term / speculative
- ▶ Growing segment with democratised access

5. Currency Pairs, Quotes & Conventions

Major Pairs (G10 — ~85% of volume)

EUR/USD	Euro / US Dollar	"Fiber" — most traded pair
USD/JPY	US Dollar / Yen	"Gopher" — Asia benchmark
GBP/USD	Pound / US Dollar	"Cable" — oldest pair
USD/CHF	US Dollar / Swiss Franc	"Swissie" — safe haven
AUD/USD	Aussie / US Dollar	Commodity-linked
USD/CAD	US Dollar / Loonie	Oil-correlated
NZD/USD	Kiwi / US Dollar	Carry-trade favourite

Key Conventions

Pip: Smallest price increment

- Most pairs: 0.0001 (4th decimal)
- JPY pairs: 0.01 (2nd decimal)

Lot Sizes:

- Standard: 100,000 base units
- Mini: 10,000 | Micro: 1,000

Cross Pairs: Pairs without USD

- EUR/GBP, EUR/JPY, GBP/CHF
- Derived from USD legs

Exotic Pairs: EM currencies

- USD/TRY, USD/ZAR, USD/BRL
- Wider spreads, less liquid

Settlement: T+2 standard (spot)

- T+1 for USD/CAD, USD/TRY

Part II

Cash FX Instruments

6. FX Spot — Mechanics & Settlement

What is an FX Spot Trade?

- ▶ An agreement to buy/sell a currency pair at the prevailing market rate
- ▶ Standard settlement: T+2 (two business days after trade date)
- ▶ The most straightforward FX instrument — represents ~28% of turnover
- ▶ Quoted as bid/ask with very tight spreads for major pairs (0.5–2 pips)
- ▶ Execution venues: interbank (EBS, Reuters), ECNs, single-dealer platforms
- ▶ Increasingly electronic — ~70% of spot volume is e-traded

Worked Example

Trade: Buy 10M EUR/USD Spot

Rate: 1.0850

Date: Monday, March 24

Settlement: Wednesday, March 26 (T+2)

Cash Flows on Settlement:

- ▶ You RECEIVE: €10,000,000
- ▶ You PAY: \$10,850,000

If rate moves to 1.0900 before settlement:

- ▶ Your EUR is now worth \$10,900,000
- ▶ Mark-to-Market P&L: +\$50,000

$$\begin{aligned}\text{P\&L} &= \text{Notional} \times (\text{Current} - \text{Entry}) \\ &= 10\text{M} \times (1.0900 - 1.0850) \\ &= 10\text{M} \times 0.0050 = \$50,000\end{aligned}$$

7. FX Forwards — Pricing & Applications

What is an FX Forward?

- ▶ A binding agreement to exchange currencies at a future date at a rate agreed today
- ▶ Settlement: any date beyond T+2 (standard tenors: 1W, 1M, 3M, 6M, 1Y, 2Y+)
- ▶ Customisable notional, tenor, and settlement terms (OTC)
- ▶ No upfront premium — both parties are obligated to settle
- ▶ Primary hedging tool for known future FX exposures

Forward Rate Formula (Covered Interest Rate Parity)

$$F = S \times (1 + r_{\text{quote}} \times T) / (1 + r_{\text{base}} \times T)$$

Where:

S = Current spot rate

r_quote = Interest rate of quote currency

r_base = Interest rate of base currency

T = Time to maturity (in years)

$$\text{Forward Points} = F - S$$

If $r_{\text{quote}} > r_{\text{base}} \rightarrow$ Forward points are POSITIVE (premium)

If $r_{\text{quote}} < r_{\text{base}} \rightarrow$ Forward points are NEGATIVE (discount)

Worked Example — 6M EUR/USD Forward

Given:

Spot EUR/USD: 1.0850

USD 6M rate: 5.25% (quote currency)

EUR 6M rate: 3.75% (base currency)

T = 0.5 years

Forward Rate Calculation:

$$F = 1.0850 \times (1 + 0.0525 \times 0.5) / (1 + 0.0375 \times 0.5)$$

$$F = 1.0850 \times 1.02625 / 1.01875$$

$$F = 1.0850 \times 1.00736$$

$$F = 1.0930$$

$$\text{Forward Points} = 1.0930 - 1.0850 = +80 \text{ pips}$$

Interpretation:

USD rates > EUR rates $\rightarrow$ EUR trades at a
FORWARD PREMIUM (costs more USD in future)

Use Case: European exporter receiving \$10M in
6 months sells USD/buys EUR at 1.0930 to lock
in guaranteed EUR proceeds.

8. FX Swaps — Structure & Use Cases

- ▶ An FX Swap = simultaneous Spot + Forward in opposite directions (same notional, same counterparty)
- ▶ Most traded FX instrument — accounts for ~51% of total FX turnover
- ▶ The swap rate (swap points) is the difference between the forward and spot rate
- ▶ NOT the same as a Cross-Currency Swap (CCS) — FX Swap has no periodic interest payments



Primary Use Cases

- ▶ Short-term funding: borrow in one currency, lend in another
- ▶ Rolling hedges: extend a maturing forward position
- ▶ Cash management: convert surplus in one currency temporarily
- ▶ Central banks: manage reserves and provide USD liquidity (swap lines)

FX Swap vs Cross-Currency Swap vs FX Forward

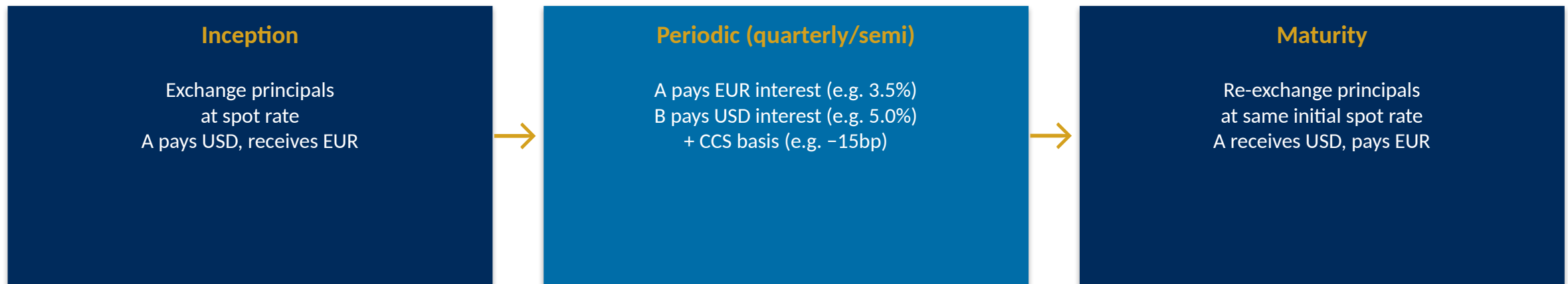
FX Swap: 2 legs (spot+fwd), no interim payments, short-dated

CCS: Periodic interest exchanges, principal exchange, long-dated

Forward: Single future exchange, one settlement date

9. Cross-Currency Swaps (CCS)

- ▶ A Cross-Currency Swap (CCS) involves exchanging interest payments in two different currencies
- ▶ Typically includes principal exchange at inception and maturity (at initial spot rate)
- ▶ Tenors are longer — typically 1Y to 30Y+ (vs FX Swaps which are short-dated)
- ▶ Can be Fixed-Fixed, Fixed-Float, or Float-Float
- ▶ The CCS basis spread reflects supply/demand for borrowing in each currency



10. Non-Deliverable Forwards (NDFs)

What is an NDF?

- ▶ A forward contract where the non-deliverable currency is not physically exchanged
- ▶ Settled in cash (typically USD) based on the difference between the agreed NDF rate and the fixing rate
- ▶ Used for currencies with capital controls or limited convertibility
- ▶ Fixing rate is set by an official source (e.g., central bank, EMTA survey)
- ▶ NDF market has grown rapidly — now ~\$300B/day
- ▶ Settlement = Notional $\times$ (NDF Rate – Fixing Rate) / Fixing Rate

Key NDF Currencies

CNY	Chinese Yuan	Offshore CNH exists but onshore is NDF
INR	Indian Rupee	RBI controls, large NDF market
KRW	Korean Won	Active NDF market in Singapore
TWD	Taiwan Dollar	Capital controls
BRL	Brazilian Real	Ptax fixing, large NDF volume
CLP	Chilean Peso	EM Latin America
COP	Colombian Peso	Growing NDF liquidity
PHP	Philippine Peso	Asia EM
IDR	Indonesian Rupiah	Capital controls
MYR	Malaysian Ringgit	Offshore restricted

Part III

FX Derivatives

11. Introduction to FX Derivatives

- ▶ FX Derivatives are contracts whose value is derived from underlying exchange rates
- ▶ They provide leveraged, flexible exposure to FX movements without full notional exchange
- ▶ Key purposes: hedging, speculation, yield enhancement, structuring
- ▶ Total outstanding OTC FX derivatives notional: ~\$110 trillion (BIS 2023)
- ▶ Traded OTC (bespoke terms) or on exchanges (standardised — e.g., CME FX futures & options)

FX Derivatives Product Map

Linear	Vanilla Options	Barrier Options	Digital / Binary	Exotic / Complex
Forwards Lock in future rate	Calls Right to buy base	Knock-In Activates at barrier	European Digital Pays at expiry	Asians Average rate payoff
NDFs Cash-settled fwd	Puts Right to sell base	Knock-Out Deactivates at barrier	One-Touch Pays if touched	Range Accruals Accrues in range
FX Futures Exchange-traded fwd	Straddles Long call + long put	Double Barrier Two barriers	No-Touch Pays if not touched	Quantos FX-protected payoff

12. Vanilla FX Options — Calls & Puts

FX Call Option

- ▶ Gives the buyer the RIGHT (not obligation) to BUY the base currency
- ▶ Buyer pays an upfront premium; seller (writer) receives it
- ▶ European-style: exercisable only on expiry date
- ▶ American-style: exercisable anytime up to and including expiry

Payoff at Expiry:

$$\text{Call Payoff} = \max(\text{Spot} - \text{Strike}, 0) \times \text{Notional}$$

Example: EUR/USD Call, $K=1.0900$, Notional=€10M

If spot at expiry = 1.1100:

$$\text{Payoff} = (1.1100 - 1.0900) \times 10\text{M} = \$200,000$$

If spot at expiry = 1.0800:

$$\text{Payoff} = \max(1.0800 - 1.0900, 0) = \$0 \text{ (expires worthless)}$$

Breakeven = Strike + Premium paid

$$\text{If premium} = \$0.0045 \text{ per EUR} \rightarrow \text{BE} = 1.0945$$

FX Put Option

- ▶ Gives the buyer the RIGHT (not obligation) to SELL the base currency
- ▶ Used to hedge against depreciation of the base currency
- ▶ Premium depends on: strike, spot, volatility, time, rates
- ▶ In FX: a EUR put is simultaneously a USD call

Payoff at Expiry:

$$\text{Put Payoff} = \max(\text{Strike} - \text{Spot}, 0) \times \text{Notional}$$

Example: EUR/USD Put, $K=1.0800$, Notional=€10M

If spot at expiry = 1.0600:

$$\text{Payoff} = (1.0800 - 1.0600) \times 10\text{M} = \$200,000$$

If spot at expiry = 1.0900:

$$\text{Payoff} = \max(1.0800 - 1.0900, 0) = \$0 \text{ (expires worthless)}$$

Breakeven = Strike - Premium paid

$$\text{If premium} = \$0.0040 \text{ per EUR} \rightarrow \text{BE} = 1.0760$$

12b. Option Moneyness, Styles & Premium

Moneyiness

In-The-Money (ITM)

Call: Spot > Strike | Put: Spot < Strike

Has intrinsic value — would be exercised

At-The-Money (ATM)

Spot ≈ Strike (or ATM-Forward)

No intrinsic value — maximum time value

Out-of-The-Money (OTM)

Call: Spot < Strike | Put: Spot > Strike

No intrinsic value — cheaper premium

Option Premium = Intrinsic Value + Time Value

- ▶ Intrinsic: $\max(\text{Spot} - K, 0)$ for call | $\max(K - \text{Spot}, 0)$ for put
- ▶ Time Value: decreases as expiry approaches (theta decay)
- ▶ ATM options have maximum time value; deep ITM/OTM have minimal

FX Option Quoting Conventions

FX options are quoted differently from equity options:

Volatility (Delta) Conventions:

- ▶ ATM: quoted as ATM-DNS (delta-neutral straddle)
- ▶ 25-delta: 25Δ Risk Reversal & Butterfly
- ▶ 10-delta: 10Δ Risk Reversal & Butterfly

Risk Reversal (RR) = Call vol – Put vol

- ▶ Measures skew (market directional bias)
- ▶ Positive RR → calls more expensive → bullish bias

Butterfly (BF) = (Call vol + Put vol)/2 – ATM vol

- ▶ Measures curvature / smile steepness
- ▶ Higher BF → fatter tails, more vol for OTM options

Premium can be quoted in:

- ▶ % of notional (base or quote currency)
- ▶ Pips per unit of base currency
- ▶ Absolute amount in settlement currency

13. Option Pricing — Black-Scholes-Garman-Kohlhagen Model

The Garman-Kohlhagen Model (1983) — Extension of Black-Scholes for FX

$$\text{Call} = S \cdot e^{(-r_b \cdot T)} \cdot N(d_1) - K \cdot e^{(-r_q \cdot T)} \cdot N(d_2)$$

$$\text{Put} = K \cdot e^{(-r_q \cdot T)} \cdot N(-d_2) - S \cdot e^{(-r_b \cdot T)} \cdot N(-d_1)$$

$$d_1 = [\ln(S/K) + (r_q - r_b + \sigma^2/2) \cdot T] / (\sigma \cdot \sqrt{T})$$

$$d_2 = d_1 - \sigma \cdot \sqrt{T}$$

Model Parameters

S = Current spot rate

K = Strike price

T = Time to expiry (years)

σ = Implied volatility (annualised)

r_b = Base currency risk-free interest rate

r_q = Quote currency risk-free interest rate

$N(\cdot)$ = Cumulative standard normal distribution

Key Assumptions & Limitations

Assumptions:

- ▶ Log-normal distribution of spot returns
- ▶ Constant volatility over option life
- ▶ Constant risk-free rates
- ▶ No transaction costs or taxes
- ▶ Continuous hedging possible

Limitations:

- ▶ Volatility is NOT constant (smile/skew exists)
- ▶ Fat tails in FX returns (more extremes than normal)
- ▶ Interest rates are stochastic, not constant
- ▶ Jumps in spot (e.g., SNB event 2015)

14. The Greeks — Sensitivity Measures

Δ Delta

Sensitivity of option price to a change in spot rate

- Ranges from 0 to 1 (calls) or -1 to 0 (puts)
- ATM option $\approx$ 0.50 delta
- Also used as probability proxy for finishing ITM
- Hedged by trading spot (delta-hedging)
- FX convention: quote strikes by delta (25 Δ , 10 Δ)

Γ Gamma

Rate of change of delta w.r.t. spot

- Highest for ATM options near expiry
- Gamma risk = hedging cost from frequent rebalancing
- Long options = long gamma (benefit from moves)
- Short options = short gamma (hurt by moves)
- Gamma P&L $\approx \frac{1}{2} \times \Gamma \times (\Delta S)^2$

v Vega

Sensitivity of option price to implied volatility

- Highest for ATM, long-dated options
- Long options = long vega (benefit from vol rising)
- Key risk for FX options desks
- Vega P&L = Vega $\times \Delta\sigma$
- Vega is quoted in \$ per 1% vol move

Θ Theta

Time decay — loss of value as expiry approaches

- Always negative for long options
- Accelerates as expiry nears (esp. ATM)
- Theta-Gamma trade-off: long gamma = pay theta
- Weekend theta: 3 days on Friday close
- Theta = daily cost of holding an option

ρ Rho

Sensitivity to interest rate changes

- In FX: two rhos — one per currency rate
- Less significant for short-dated options
- Important for long-dated CCS & options
- Higher base rate $\rightarrow$ lower call value
- Higher quote rate $\rightarrow$ higher call value

14b. Greeks Interactions & Dynamic Hedging

Key Greek Relationships

- ▶ Delta-Gamma: Gamma is the rate of change of delta. High gamma means delta changes rapidly → more frequent rebalancing needed
- ▶ Theta-Gamma Trade-off: You cannot be long gamma (profit from moves) without paying theta (time decay). This is fundamental.
- ▶ Vega-Gamma: Both peak for ATM options but vega peaks for long-dated, gamma peaks for short-dated. Different risk profiles.
- ▶ Vanna ($d\Delta/dVol$): How delta changes when vol changes. Critical for managing skew/smile risk.
- ▶ Volga ($dVega/dVol$): Convexity in vega. Important for OTM options and drives smile pricing adjustments.

Delta Hedging in Practice

Goal: Neutralise directional (spot) risk

Process:

1. Sell EUR/USD call → initial delta = -0.50
2. Buy $0.50 \times$ Notional EUR spot to hedge
3. As spot moves, delta changes (gamma effect)
4. Rebalance spot position continuously

Delta-Hedging P&L:

- ▶ Realised vol > Implied vol → Loss for seller
- ▶ Realised vol < Implied vol → Profit for seller
- ▶ Hedging P&L = $f(\text{Gamma, actual vs implied vol})$

Practical Considerations:

- ▶ Hedging frequency: continuous (theory) vs discrete (practice — every \$X move or N times/day)
- ▶ Transaction costs erode hedging P&L
- ▶ Gamma scalping: deliberately rebalancing to capture profits from large spot moves

15. FX Volatility — Smile, Skew & Surface

Understanding FX Volatility

- ▶ Implied Volatility (IV): market's expectation of future realised volatility
- ▶ IV varies by strike (smile/skew) and by tenor (term structure)
- ▶ Together they form the 3D Volatility Surface

The Volatility Smile:

- ▶ IV is NOT constant across strikes (unlike BS assumption)
- ▶ OTM puts and calls trade at higher IV than ATM
- ▶ Driven by: fat tails, jump risk, supply/demand imbalances

Skew (Risk Reversal):

- ▶ Asymmetry in the smile — one side steeper than the other
- ▶ EUR/USD: typically put-skew (downside protection more expensive)
- ▶ USD/JPY: call-skew (upside risk from rapid yen weakening)

Term Structure:

- ▶ Short-term vol: event-driven, spiky (elections, NFP, ECB)
- ▶ Long-term vol: mean-reverting, more stable
- ▶ Vol tends to be higher around macro events (FOMC, CPI, etc.)

Building the Vol Surface

Market Quotes (per tenor):

- ▶ ATM-DNS (delta-neutral straddle) volatility
 - ▶ 25Δ Risk Reversal (25Δ Call vol – 25Δ Put vol)
 - ▶ 25Δ Butterfly ((25Δ Call + 25Δ Put)/2 – ATM)
 - ▶ 10Δ Risk Reversal
 - ▶ 10Δ Butterfly
- 5 quotes × N tenors = full surface

Standard Tenors:

O/N, 1W, 2W, 1M, 2M, 3M, 6M, 9M, 1Y, 2Y, 3Y, 5Y

Interpolation Methods:

- ▶ SABR model (stochastic alpha-beta-rho)
- ▶ Vanna-Volga method
- ▶ SVI (Stochastic Vol Inspired) parameterization
- ▶ Cubic spline on delta space

Typical EUR/USD ATM Vols (example):

1M: 7.5% | 3M: 8.2% | 6M: 8.8% | 1Y: 9.3%

16. Barrier Options — Knock-In / Knock-Out

- ▶ Barrier options are path-dependent options that activate (knock-in) or deactivate (knock-out) if spot touches a predefined barrier level
- ▶ Always cheaper than equivalent vanilla options (conditional payoff → lower premium)
- ▶ Widely used in FX for cost-effective hedging and structured products

Down-and-In (DI)

Barrier BELOW spot
Option activates if spot falls to barrier

Use: Cheap call that only exists if market drops first

Example: EUR/USD spot 1.0850
DI Call, K=1.0900, Barrier=1.0600
Call only activates if EUR drops to 1.0600

Down-and-Out (DO)

Barrier BELOW spot
Option deactivates if spot falls to barrier

Use: Hedge that disappears if market moves too far

Example: EUR/USD spot 1.0850
DO Call, K=1.0900, Barrier=1.0600
Call dies if EUR drops to 1.0600

Up-and-In (UI)

Barrier ABOVE spot
Option activates if spot rises to barrier

Use: Cheap put protection that activates on rally

Example: EUR/USD spot 1.0850
UI Put, K=1.0800, Barrier=1.1200
Put only activates if EUR rises to 1.1200

Up-and-Out (UO)

Barrier ABOVE spot
Option deactivates if spot rises to barrier

Use: Cheaper hedge with upper cap on protection

Example: EUR/USD spot 1.0850
UO Put, K=1.0800, Barrier=1.1200
Put dies if EUR rises to 1.1200

16b. Barrier Options — Pricing, Risk & Variations

Pricing & Risk Characteristics

- ▶ In-Out Parity: Knock-In + Knock-Out = Vanilla (same strike/barrier)
 - Used for pricing and risk management
- ▶ Barrier options have discontinuous delta near the barrier level
 - Pin risk: delta jumps dramatically as spot approaches barrier
 - Creates hedging challenges for dealers
- ▶ Vega profile differs from vanilla — can change sign near barrier
- ▶ Barrier monitoring: Continuous (standard) vs Discrete (daily fix)
 - Discrete barriers are cheaper (less likely to be hit)
- ▶ Rebate: Some barrier options pay a fixed amount if knocked out
 - Compensates buyer for loss of option

Barrier Variations

Double Barrier (Double Knock-Out):

- ▶ Two barriers — upper and lower
- ▶ Option dies if either barrier is touched
- ▶ Significantly cheaper than single barrier
- ▶ Popular in range-bound markets

Window Barrier:

- ▶ Barrier only monitored during a specific window
- ▶ E.g., barrier active only in last month of 3M option

Partial Barrier:

- ▶ Barrier monitored for part of the option's life

Step / Fading Barrier:

- ▶ Barrier level changes over time
- ▶ Or option partially knocks out as spot lingers near barrier

Parisian Barrier:

- ▶ Must stay beyond barrier for N days to trigger
- ▶ Reduces manipulation / whipsaw risk

17. Digital / Binary Options

What are Digital Options?

- ▶ Pay a fixed amount if a condition is met — otherwise pay nothing
- ▶ Also known as Binary Options or Bet Options
- ▶ Payoff is discontinuous (all-or-nothing)
- ▶ Priced as the limit of a tight call/put spread
- ▶ Delta is very large near strike at expiry → hedging challenge
- ▶ Common in FX for event-driven hedging and structured products

European Digital Call

Pays fixed amount if Spot > Strike at expiry
Payoff = Fixed $\times 1\{S_T > K\}$
Example: Pays \$100K if EUR/USD > 1.10 at expiry

European Digital Put

Pays fixed amount if Spot < Strike at expiry
Payoff = Fixed $\times 1\{S_T < K\}$
Example: Pays \$100K if EUR/USD < 1.05 at expiry

One-Touch

Pays fixed amount if Spot EVER touches the barrier
Path-dependent — American-style digital
Example: Pays \$200K if EUR/USD ever reaches 1.12
Cheaper if barrier is far from spot

No-Touch

Pays fixed amount if Spot NEVER touches the barrier
Inverse of one-touch: 1-Touch + No-Touch = PV(Fixed)
Example: Pays \$150K if EUR/USD never reaches 1.12
Popular for yield enhancement

18. Advanced Exotic FX Structures

Asian / Average Rate Options

- ▶ Payoff based on AVERAGE spot rate over a period (not final spot)
- ▶ Average < volatility of single observation → cheaper than vanilla
- ▶ Arithmetic average (standard) vs Geometric average (closed-form)
- ▶ Uses: hedging regular FX flows (monthly revenues), commodity-linked FX
- ▶ Fixing schedule: daily, weekly, or monthly observations
- ▶ Payoff = $\max(\text{Average} - K, 0) \times \text{Notional}$ [for average rate call]

Range Accruals

- ▶ Accrues payoff for each day that spot stays within a defined range
- ▶ Payout = $(\text{Fixed Coupon}) \times (\text{Days In Range} / \text{Total Days})$
- ▶ Yield enhancement: investor earns above-market coupon if range holds
- ▶ Dual range accruals: two underlying rates must both be in range
- ▶ Popular in structured deposit and note markets
- ▶ Risk: zero payout if spot breaks range frequently

Quanto Options

- ▶ Payoff is in a DIFFERENT currency from the underlying
- ▶ Eliminates FX risk on foreign asset returns
- ▶ Example: Option on Nikkei 225, but payoff in USD (not JPY)
- ▶ Requires modelling correlation between asset and FX rate
- ▶ Quanto adjustment: drift adjustment for correlation
- ▶ Used by global asset managers to isolate pure asset returns

Target Redemption Forwards (TARFs)

- ▶ Series of FX forwards with an embedded knock-out on cumulative profit
- ▶ Once cumulative gain reaches 'target', all remaining forwards cancel
- ▶ Leveraged: downside notional typically 2x upside notional
- ▶ Very popular in Asia and Latin America for corporate hedging
- ▶ Significant risk: unlimited downside if spot moves against you
- ▶ Regulatory scrutiny due to complex risk profile

18b. More Exotic Structures

Accumulator / Decumulator

- ▶ Obligates buyer to purchase (accumulator) or sell (decumulator) fixed amounts at each fixing if spot is favourable
- ▶ If spot moves against, buyer accumulates at 2x or 3x rate (leverage)
- ▶ KO feature: if spot rallies past barrier, deal terminates
- ▶ Very popular in Asia; known as 'I kill you later' due to risk

Best-Of / Worst-Of Options

- ▶ Payoff linked to best or worst performing currency in a basket
- ▶ Worst-of: cheaper than individual options (correlation benefit)
- ▶ Best-of: more expensive (captures all upside)
- ▶ Correlation is the key risk driver — hard to hedge
- ▶ Used in multi-currency structured products

Compound Options (Option on Option)

- ▶ The right to buy/sell another option at a future date
- ▶ Four types: Call-on-Call, Call-on-Put, Put-on-Call, Put-on-Put
- ▶ Used when hedging need is uncertain (e.g., contingent M&A)
- ▶ Cheaper upfront than buying the underlying option outright
- ▶ Two strike prices and two expiry dates

Chooser Options

- ▶ Holder can choose whether the option is a call or put at a specified date
- ▶ Flexibility premium — more expensive than either call or put alone
- ▶ Useful when direction is uncertain but volatility event is certain
- ▶ Example: pre-election hedge — choose call or put after result
- ▶ Can be decomposed into call + put with adjusted terms

Part IV

Risk Management & Practice

19. FX Risk Management — Types of Risk

Transaction Risk

Risk from FX movements between trade date and settlement

- ▶ Arises from import/export invoices in foreign currency
- ▶ Time lag between pricing a deal and cash settlement
- ▶ Directly impacts P&L — most immediate FX risk
- ▶ Example: US importer buys goods priced in EUR

Invoice: €5M due in 90 days

If EUR/USD moves from 1.08 to 1.12:

Cost increase = $5M \times (1.12 - 1.08) = \$200,000$

Hedging: Forwards, Options, Natural hedging

Translation Risk

Risk from consolidating foreign subsidiary financials

- ▶ Parent company must translate foreign sub's P&L and balance sheet
- ▶ Exchange rate at reporting date affects consolidated accounts
- ▶ Does not affect cash flow directly — but impacts reported earnings
- ▶ Example: US parent with UK subsidiary
GBP revenue looks smaller if GBP depreciates
Can cause EPS volatility even if operations are stable

Hedging: Balance sheet hedging, net investment hedges

Many firms choose NOT to hedge (non-cash impact)

Economic / Strategic Risk

Long-term competitive impact of FX on business value

- ▶ Persistent FX moves change competitive dynamics
- ▶ Strong domestic currency → exports less competitive
- ▶ Affects market share, pricing power, investment decisions
- ▶ Hardest to measure and hedge
- ▶ Example: Strong JPY in 2010-2012
Japanese automakers lost competitiveness to Korean rivals
Led to strategic shifts (overseas production)

Hedging: Operational (diversify production/sourcing)

Long-dated options, strategic currency allocation

20. Hedging Strategy Comparison — Deep Dive

Forward Hedge	Vanilla Option	Collar (Risk Reversal)	Participating Forward	Knock-Out Forward
Lock in exchange rate for future date	Buy call or put for protection	Buy put + sell call (or vice versa)	Forward + buy-back option on portion	Forward with barrier — cancels if touched
✓ Complete certainty on rate ✓ Zero premium cost ✗ No upside participation ✗ Binding obligation — cannot walk away ✗ Opportunity cost if market moves favourably	✓ Full downside protection ✓ Unlimited upside participation ✓ Flexibility — can let it expire ✗ Premium cost (can be significant) ✗ Premium is non-refundable	✓ Low or zero cost (premiums offset) ✓ Downside protection guaranteed ~ Upside capped at sold strike ✗ Still obligated on sold leg ✗ May have margin requirements	✓ Full downside protection ✓ Partial upside (e.g., 50%) ~ Moderate premium ✗ More complex than vanilla ✗ Participation ratio affects cost	✓ Better-than-market forward rate ✓ Zero or low cost ✗ Protection disappears if barrier hit ✗ Path-dependent risk ✗ Can leave you unhedged
Best For: Corporates with known, certain cash flows	Best For: Uncertain cash flows, competitive bids	Best For: Cost-sensitive corporates, budget rate hedging	Best For: Corporates wanting some upside	Best For: Risk-tolerant, view-driven hedgers

Key Principle: There is no 'best' hedging strategy — the optimal choice depends on risk appetite, cash flow certainty, cost tolerance, and market view.

21. Real-World Use Cases — Part 1

1

European Exporter — Hedging USD Receivables

Scenario: German auto manufacturer invoices \$500M in quarterly USD revenues

Challenge: EUR/USD appreciation would reduce EUR value of USD receivables

Solution: Rolling 3-month EUR/USD forward programme

- ▶ Sell USD/Buy EUR forward for 75% of expected receivables
- ▶ Leave 25% unhedged for potential EUR weakness
- ▶ Use vanilla puts for the remaining 25% (downside protection + upside)

Result: Budget rate certainty on core revenues; partial upside participation

2

US Pension Fund — Currency Overlay

Scenario: \$2B allocated to European equities, exposed to EUR depreciation

Challenge: Unhedged FX can add 5-10% annual volatility to returns

Solution: 50% passive hedge using 1M rolling forwards + 25% option overlay

- ▶ Passive component: sell EUR/buy USD 1M forwards, rolled monthly
- ▶ Active component: buy EUR/USD puts when implied vol is cheap
- ▶ Remaining 25% unhedged as long-term EUR exposure

Result: Reduced FX volatility by ~60%; hedging cost ~25bp/year

21b. Real-World Use Cases — Part 2

3

Structured Product — Dual Currency Note

Scenario: Investor seeks enhanced yield on 6-month USD deposit

Structure: Investor deposits \$1M at 8% p.a. (vs 5.25% normal rate)

- ▶ Embedded: Investor sells EUR/USD put to bank ($K = 1.0500$)
- ▶ At maturity: if EUR/USD < 1.0500, investor receives principal in EUR
- ▶ If EUR/USD ≥ 1.0500 , investor receives \$1M + enhanced coupon

Risk: If EUR drops significantly, investor receives EUR worth less than \$1M
The enhanced yield (2.75% extra) compensates for the put risk

4

EM Central Bank — Managing Currency Pressure

Scenario: EM central bank facing rapid depreciation of local currency

Tools available:

- ▶ FX reserves: direct intervention — sell USD, buy local currency
- ▶ FX swaps with Fed (swap lines): borrow USD to stabilise
- ▶ NDFs: intervene in offshore market to anchor expectations
- ▶ Interest rate hikes: make local currency attractive (carry)

Considerations: Reserve adequacy, market credibility, economic cost
Example: Turkey 2021-22, India RBI regular NDF intervention

22. FX Regulatory Landscape

Key Regulatory Frameworks

- ▶ FX Global Code (2017, updated 2021)
 - 55 principles for fair and effective FX markets
 - Voluntary adherence via Statement of Commitment
 - Covers: ethics, execution, governance, information sharing
- ▶ Dodd-Frank Act (US) / EMIR (EU) / MiFID II
 - Mandatory reporting of OTC FX derivatives
 - Central clearing for certain standardised products
 - Margin requirements for uncleared derivatives
- ▶ Basel III / SA-CCR
 - Standardised Approach for Counterparty Credit Risk
 - Higher capital charges for FX derivatives
 - Incentivises central clearing and netting
- ▶ FRTB (Fundamental Review of the Trading Book)
 - New market risk capital framework
 - Separate treatment for FX risk factors

Industry Trends & Evolution

Electronification:

- ▶ 70%+ of spot volume is electronic
- ▶ Algos dominate market-making
- ▶ ECNs, multi-dealer platforms, single-dealer APIs

FX Futures Growth:

- ▶ CME FX futures gaining share from OTC
- ▶ Capital efficiency via central clearing
- ▶ EM currencies increasingly listed

Digital / Blockchain:

- ▶ DLT-based settlement (CLS exploring)
- ▶ Stablecoins and CBDC impact on FX
- ▶ Real-time settlement potential (T+0)

Data & AI:

- ▶ Machine learning for flow prediction
- ▶ NLP for news-driven trading signals
- ▶ AI-driven optimal execution / TCA

23. Key Takeaways

- 1** **FX is the world's largest financial market**
~\$7.5 trillion daily turnover, 24-hour, decentralised OTC
- 2** **Core cash instruments**
Spot (T+2), Forwards (future rate lock), FX Swaps (51% of volume), NDFs (EM hedging)
- 3** **Forward pricing is driven by interest rate differentials**
Covered Interest Rate Parity: $F = S \times (1+r_q \cdot T) / (1+r_b \cdot T)$
- 4** **FX Options provide asymmetric risk/reward**
Pay premium for downside protection while retaining upside — calls, puts, and combinations
- 5** **Garman-Kohlhagen model is the standard FX pricing model**
Extension of Black-Scholes with two interest rates; limited by constant vol assumption
- 6** **The Greeks are essential for risk management**
Delta, Gamma, Vega, Theta, Rho — plus higher-order: Vanna, Volga
- 7** **Volatility surface captures market expectations**
Smile, skew, and term structure; quoted via ATM, RR, BF conventions
- 8** **Exotic derivatives enable tailored solutions**
Barriers, Digitals, Asians, TARFs, Accumulators — but with complex risk profiles
- 9** **Hedging strategy must match risk profile**
No one-size-fits-all; consider: certainty, cost, upside participation, complexity
- 10** **Regulation is reshaping FX markets**
FX Global Code, clearing mandates, margin rules, capital requirements

Recommended Reading & Resources



"Foreign Exchange Option Pricing" — Iain Clark

Comprehensive guide to FX option pricing with practical examples



"FX Options and Structured Products" — Uwe Wystup

Industry standard reference for FX derivatives structuring



"Options, Futures, and Other Derivatives" — John Hull

Foundational textbook covering all derivative types



"Currency Trading and Intermarket Analysis" — Ashraf Laidi

Practical guide to FX market dynamics and cross-asset relationships



Resources:

BIS Triennial Central Bank Survey

bis.org — definitive FX market statistics

FX Global Code

globalfx.org — 55 principles for FX markets

CLS Group

cls-group.com — settlement data and insights

CME FX

cmegroup.com/fx — FX futures and options data

Bloomberg / Reuters

Real-time FX market data, volatilities, analytics

Questions & Discussion

Thank You